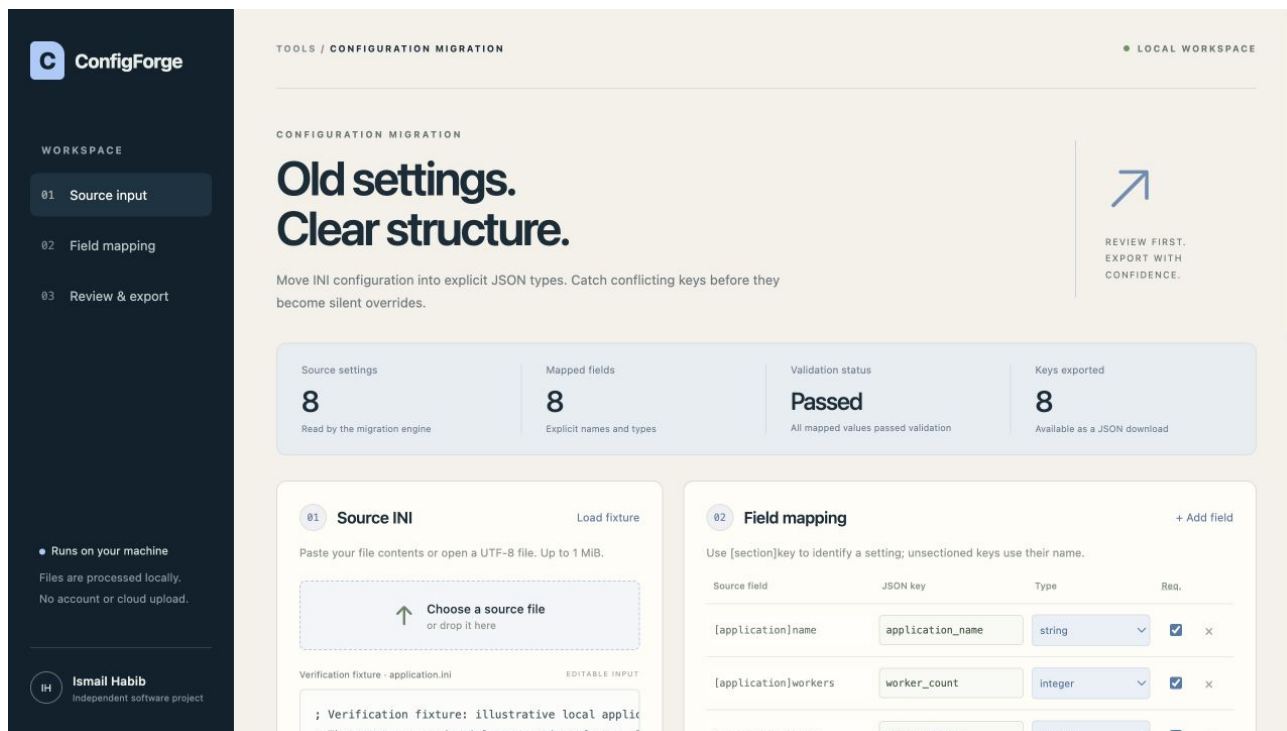


ConfigForge

INI-to-JSON configuration migration with conflict detection

Independent software project | Owner: Ismail Habib | Created September 2026



Actual application capture. Any example records are labeled verification inputs.

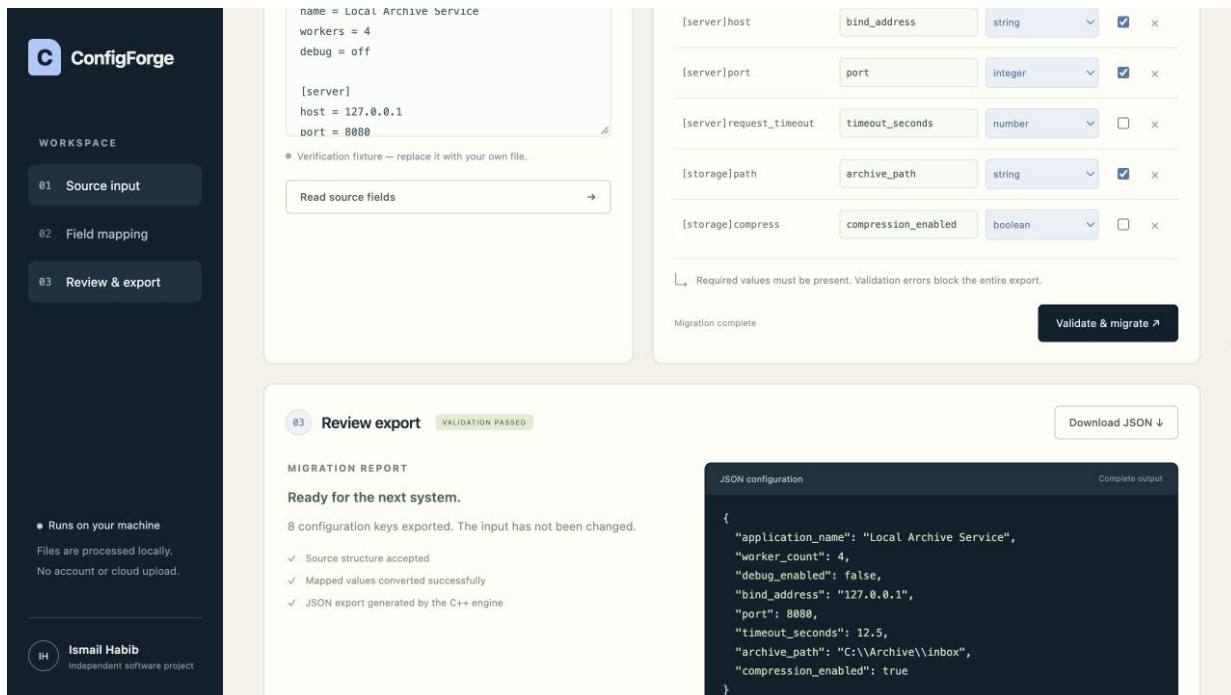
The problem

Existing applications often store settings as strings in INI files, while newer systems require typed JSON. Repeated settings can silently override values, booleans can be interpreted incorrectly, and escaping can alter directory paths. A reliable migration workflow needs to expose conversion decisions and reject ambiguous configuration before publishing an export.

What the project delivers

- C++17 INI parsing with duplicate-section and duplicate-key rejection
- Explicit section/key mapping to typed JSON configuration fields
- Required-value checks and source-line diagnostics before export
- Literal-path preservation and protected command-line output publishing

Built to be inspected.



The validated JSON preview shows the actual generated settings, including typed values and the preserved path string from the verification fixture.

What was verified

Sixteen behavioral tests passed against the compiled engine and adapter on 28 September 2026. The disclosed fixture produced eight JSON settings, including integers, a decimal timeout, booleans and a preserved Windows path string. Tests exercised duplicate settings, malformed quoting, optional and missing values, invalid types, UTF-8 and prevention of input/output overwrite. Apple Clang 17 compiled C++17 without warnings on macOS 15.7.8; Python and JavaScript syntax checks passed. Windows-path preservation was verified as string handling, not proof of Windows runtime compatibility.

How it works

Read INI settings > Declare target schema > Reject conflicts and errors > Export typed configuration

Current scope

ConfigForge is a new independent migration utility, not a rewrite of a historical customer codebase. Its documented INI dialect does not support interpolation, includes, multiline continuation, nested JSON output or silent duplicate overrides. Type validation does not establish application-specific rules, such as acceptable network-port ranges. The tool exports settings but does not deploy or apply them. Execution was verified on macOS; Windows compilation and runtime behavior remain unverified. Real adoption requires checking the source dialect and target application's configuration expectations.

Included: source code, setup guide, tests, screenshots, editable case study and Malt copy.
MIT License | Copyright 2026 Ismail Habib.